Fatemeh Maddahzadeh

Software Engineer - Backend & AI/LLM
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Profile

Backend-focused Software Engineer with 6+ years of experience in Node.js/TypeScript, microservices and cloud (Google Cloud Platform, Kubernetes). At Glue Labs I was the founding backend engineer of a production multi-tenant LLM platform: RAG on PostgreSQL/pgvector, multi-provider LLM integration, tool calling, MCP and AI agents, background jobs and deployment on GCP. Former Backend Team Leader (Scrum, code reviews, mentoring). MSc in Software Applications at Politecnico di Torino; thesis on multi-LLM systems. Available immediately.

Professional Experience

- Software Engineer at Glue Labs

Mar. 2023 - Dec. 2025 | Turin, Italy | Part-time, alongside MSc | Pupau (pupau.ai): a multi-tenant B2B platform where companies build AI assistants and agents on their own knowledge bases.

- **Founding backend engineer:** built the core platform alone for its first ~6 months (data model, ingestion, RAG, conversations, deployment), then kept ownership of the AI core as the team grew to up to 7 developers (~1,650 commits, 2nd-highest contributor)
- Built modular services with NestJS, TypeORM and RxJS: multi-tenant access with API keys, background jobs on Google Cloud Tasks and Cloud Scheduler, and per-call cost tracking of LLM and embedding usage
- **Built end-to-end RAG:** document and web ingestion, tiktoken-based chunking, async embedding, PostgreSQL + pgvector similarity search (IVFFlat) and versioned chunks for zero-downtime re-indexing
- Designed a multi-provider LLM and embedding layer (OpenAI, Azure OpenAI, Anthropic, Google Gemini, Vertex AI, Groq, Mistral) behind one interface, with a factory for provider selection and load balancing
- Created the actor-based AI agent framework (later grown by the team to ~30 actors: web search, code interpreter, browser automation, database) for multi-step pipelines, with tool calling, an MCP client and approval before tools run
- **Ran it in production:** Server-Sent Events streaming, LLM guardrails, LLM call logs and Prometheus metrics; set up the first staging and production environments on GCP, later run on GKE and App Engine with versioned DB migrations

- Backend Team Leader at Lamasoo

Jan. 2022 - Sep. 2022 | Tehran, Iran | Online channel manager for the hospitality industry, serving hotels and agencies.

- **Led a team of 3 backend developers:** hiring, mentoring, task assignment and progress reviews
- Planned sprints and technical tasks using Scrum, running daily stand-ups and review meetings
- Conducted code reviews to ensure code quality and compliance with technical and client requirements
- Designed scalable microservices and oversaw system changes and performance
- Worked with the product team on priorities and fast resolution of reported bugs

- Software Engineer at Lamasoo

Jun. 2020 - Jan. 2022 | Tehran, Iran

- Designed and implemented multiple microservices in Node.js/TypeScript
- Refactored the codebase towards a microservices architecture, improving maintainability
- Developed RESTful APIs and worked with front-end developers on integration
- Optimized database queries and performance, writing efficient, reusable and testable code

- Software Engineer at Linkap

Feb. 2019 - Jun. 2020 | Tehran, Iran | IoT start-up in the smart home industry.

- Developed the Node.js backend of a smart home application
- Integrated IoT devices (LoRaWAN, Zigbee) in real time, with dynamic storage of device data in MongoDB
- Contributed to the system architecture design and wrote technical documentation (API and database design)

Education

- Politecnico di Torino

Sep. 2022 - Nov. 2026 | Turin, Italy | MSc in Software Applications

- GPA 25/30
- **Thesis:** Handoff-Aware Routing and Drift Detection for Multi-LLM Systems. Built a Python middleware on LiteLLM that detects mid-conversation model switches; compared 4 mitigation strategies across 400 trials with 5 LLMs. The best strategy raised the full-pass rate from 67% to 74%

- University of Tehran

Sep. 2016 - Feb. 2021 | Tehran, Iran | BSc in Computer Engineering (Software Engineering)

- **GPA 16.5/20 (last two years:** 17.36/20); ranked the top university in Iran by U.S. News
- Ranked 61st in the national university entrance exam

Selected Project

- Mismatch Desk | github.com/fatemeh76m/mismatch-desk | Invoice discrepancy review | Portfolio project

- Developed FastAPI endpoints and a LangGraph workflow with deterministic supplier, item, quantity and price checks across invoices, purchase orders and delivery records.
- Added PostgreSQL/Alembic persistence for review cases and append-only human decisions, text-based PDF extraction with pypdf, and a Next.js review UI.
- OpenRouter structured extraction includes exact source quotes and flags missing or ambiguous fields for human review. Uses synthetic data; no automatic approval or payment.

Technical Skills

- **Languages:** TypeScript, JavaScript (Node.js), Python, C++ (university projects)
- **Backend:** FastAPI, pypdf, NestJS, Express.js, TypeORM, Sequelize, Mongoose, RxJS, REST APIs, Server-Sent Events, microservices, background jobs and message queues
- **Frontend:** HTML, CSS, React, Next.js
- **AI / LLM:** RAG, embeddings, pgvector, tool calling, MCP, AI agents, LLM guardrails, prompt engineering, LiteLLM; OpenAI, Azure OpenAI, Anthropic, Gemini, Vertex AI, Groq, Mistral, LangGraph, OpenRouter, evidence-grounded structured extraction, human-in-the-loop review
- **Databases:** PostgreSQL, MongoDB, Redis, SQL, Alembic migrations
- **Cloud & DevOps:** Google Cloud Platform (GKE, App Engine, Cloud Build, Cloud Tasks, Cloud Storage), Kubernetes, Docker, Prometheus, Git, Linux
- **Messaging & IoT:** RabbitMQ, MQTT, LoRaWAN, Zigbee
- **Other:** Scrum, Jira, design patterns, MVC
- **AI-assisted development:** Codex, Cursor

Languages

Persian: native | English: Advanced | Italian: Intermediate